Game Architecture Design Document

1. Scope

This document describes a modular, engine-portable architecture for a character-driven action RPG with:

- Multi-domain character control (Traversal/Posture/CombatMode/Action)
- Input → Intent resolution + arbitration
- Stealth, interaction opportunities, and social actions ("Reason") during combat
- Phase-based world variants (Autumn/Winter/Spring) with per-phase deltas and selective cross-phase `WorldObjectId` continuity
- A reactive world director ("GameMaster") that schedules events outside direct player control
- Persistence designed to scale (JSON first; SQLite/chunked later)

The architecture is designed to:

- Avoid state explosion
- Avoid mixed concerns (Unity-specific code stays in adapters)
- Make the Player and NPCs "chess pieces" with stable outward APIs

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2. Design Principles

2.1 Hard boundaries

- Core gameplay modules **must not** depend on engine/runtime APIs (Unity/Stride).
- Engine host layers convert between engine objects and **pure snapshots/outputs**.

2.2 Durable vs transient

- Save only durable model state (stats, quests, equipment, reputation, world deltas).
- Do not persist transient runtime state (FSM instances, animation progress, input buffers) unless explicitly required.

2.3 Ownership

Each domain owns a small set of decisions and state variables:

- Traversal: grounded/airborne/swim/climb semantics
- Posture: stance + collider profile + baseline stealth multipliers
- CombatMode: sheathed/unsheathed, ranged vs melee readiness, aiming mode
- Action: committed action execution, stamina costs, cooldowns, buffering, cancel windows

2.4 Input resolution is separate

Unity Input System handles *physical mapping* and interactions (tap/hold/chords).
Game logic resolves *meaning in context* and arbitrates conflicts.

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3. Repository / Solution Layout

```
/docs
Architecture.md
PlayerArchitecture.md
Persistence.md
WorldPhases.md
GameMaster.md

/src
Game.SharedKernel
Game.Content
Game.World.Core
Game.Characters.Core
Game.Quests.Core
Game.GameMaster.Core
Game.Persistence.Core
Game.Persistence.Json
Game.Adapter.Unity
Game.UnityHost
/tests
Game.Characters.Core.Tests
Game.World.Core.Tests
Game.GameMaster.Core.Tests
Game.Persistence.Core.Tests
```

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4. Core Module Boundaries

4.1 Game.SharedKernel

- IDs: `EntityId`, `FactionId`, `QuestId`, `SkillId`, `WorldObjectId`, `CellId`
- Math primitives: `Vec2`, `Vec3`, `Angle`
- Messaging primitives: `DomainEvent`, `IEventBus`

4.2 Game.Content

Static definitions (data-driven):

- `SkillDefinition`, `WeaponDefinition`, `QuestDefinition`
- Phase world catalogs (which content belongs to Autumn/Winter/Spring)

4.3 Game.World.Core

- `WorldPhase`, `WorldClock`
- World partitioning (cells/chunks), streaming policy
- Read/write ports: `IWorldQuery`, `IWorldCommand`
- Deltas: per-phase, per-cell
- `WorldObjectState` for selective cross-phase continuity
- World events bus

4.4 Game.Characters.Core

- `CharacterPieceBase`, `PlayerPiece`, `NpcPiece`
- Durable `CharacterModel`: stats, reputation, equipment, skill loadouts
- Input snapshot → Intent resolution → arbitration
- Domains: Traversal/Posture/CombatMode/Action
- Systems: stealth telemetry + opportunities, interaction resolver, social attempt hooks
- Output commands + events

4.5 Game.Quests.Core

- `QuestJournal` and quest instance states
- Phase gating, expiry rules, phase transition policies
- Quest events

4.6 Game.GameMaster.Core

- Reactive world director
- Rules, task scheduling, condition triggers
- Persists its own agenda and world metrics

4.7 Game.Persistence.Core (+ Json)

- Save models: global + per-phase cell deltas + worldobject state + player + quests + GM state
- Versioning + migrations
- Storage port `ISaveStore`, serializer port `ISaveSerializer`

4.8 Adapters / Hosts

- Unity adapters build snapshots and apply outputs
- Unity host composes systems and wires DI

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5. The Chess Piece Concept

A “piece” (player/NPC/monster) is a unit with:

- A stable **read model** interface for the rest of the world
- A narrow **command port** for effects applied to it (damage/status/impulses/social offers)
- An event stream emitted outward
- Internal modules that are private (no external access)

5.1 Outward API contracts

- `ICharacterReadModel`:
  - identity + faction
  - kinematics: position/forward/velocity
  - modes: traversal, posture, combat mode, aiming
  - action: locked/current action tag
  - stealth: noise/visibility emission
  - resources: health/stamina
- `ICharacterCommandPort`:
  - ApplyDamage / ApplyImpulse / ApplyStatus
  - ReceiveInteractionOffer (optional)

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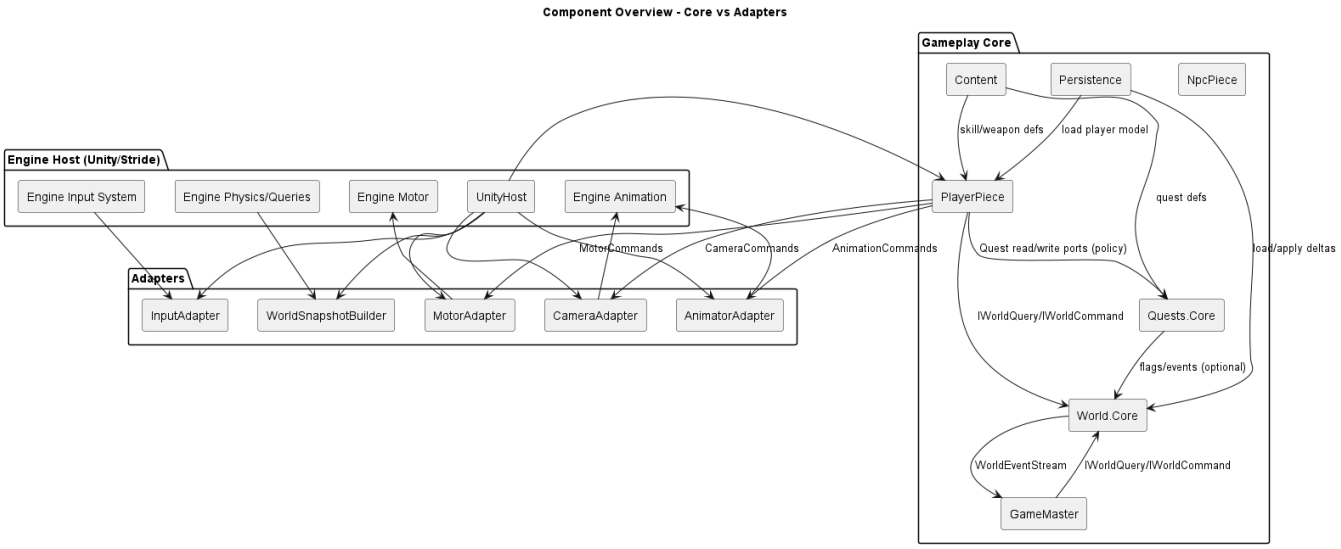
6. Diagrams

6.1 Component overview

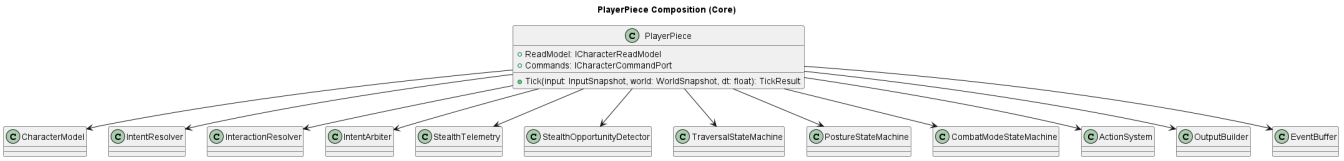



```
@startuml
skinparam componentStyle rectangle
title Component Overview - Core vs Adapters
```

```
package "Engine Host (Unity/Stride)" {
  [UnityHost] as UH
  [Engine Input System] as EIS
  [Engine Physics/Queries] as EPQ
  [Engine Animation] as EA
  [Engine Motor] as EM
}

package "Adapters" {
  [InputAdapter] as IA
  [WorldSnapshotBuilder] as WSB
  [MotorAdapter] as MA
  [AnimatorAdapter] as AA
  [CameraAdapter] as CA
}

package "Gameplay Core" {
  [PlayerPiece] as PP
  [NpcPiece] as NP
  [World.Core] as WC
  [Quests.Core] as QC
  [GameMaster] as GM
  [Persistence] as PS
  [Content] as CT
}

EIS --> IA
EPQ --> WSB

UH --> IA
UH --> WSB
UH --> PP
UH --> MA
UH --> AA
UH --> CA

PP --> WC : IWorldQuery/IWorldCommand
PP --> QC : Quest read/write ports (policy)
GM --> WC : IWorldQuery/IWorldCommand
QC --> WC : flags/events (optional)
PS --> WC : load/apply deltas
PS --> PP : load player model
CT --> PP : skill/weapon defs
CT --> QC : quest defs
WC --> GM : WorldEventStream

PP --> MA : MotorCommands
PP --> AA : AnimationCommands
PP --> CA : CameraCommands
MA --> EM
AA --> EA
CA --> EA
```

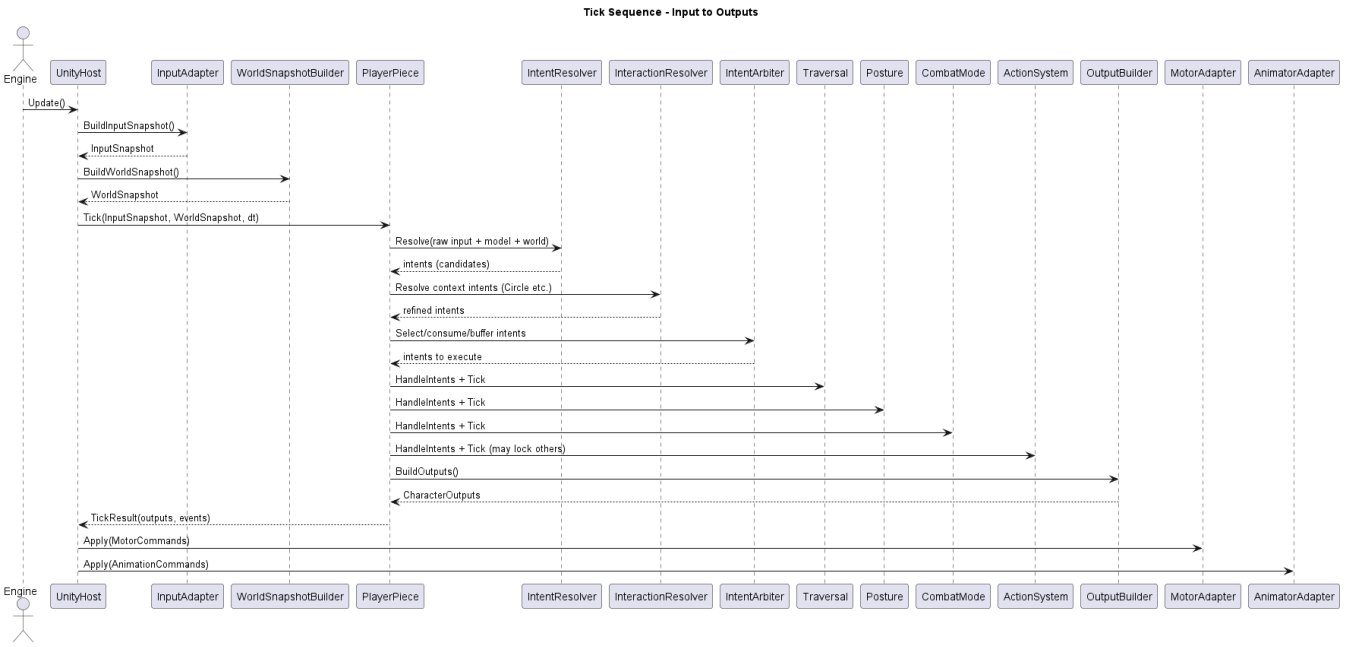
```
@enduml
```

6.2 PlayerPiece internal composition




```
@startuml
title PlayerPiece Composition (Core)

class PlayerPiece {
  +Tick(input: InputSnapshot, world: WorldSnapshot, dt: float): TickResult
  +ReadModel: ICharacterReadModel
  +Commands: ICharacterCommandPort
}

class CharacterModel
class IntentResolver
class InteractionResolver
class IntentArbiter
class StealthTelemetry
class StealthOpportunityDetector

class TraversalStateMachine
class PostureStateMachine
class CombatModeStateMachine
class ActionSystem

class OutputBuilder
class EventBuffer

PlayerPiece --> CharacterModel
PlayerPiece --> IntentResolver
PlayerPiece --> InteractionResolver
PlayerPiece --> IntentArbiter
PlayerPiece --> StealthTelemetry
PlayerPiece --> StealthOpportunityDetector

PlayerPiece --> TraversalStateMachine
PlayerPiece --> PostureStateMachine
PlayerPiece --> CombatModeStateMachine
PlayerPiece --> ActionSystem

PlayerPiece --> OutputBuilder
PlayerPiece --> EventBuffer

@enduml
```

6.3 Frame update sequence (input → intent → domains → outputs)

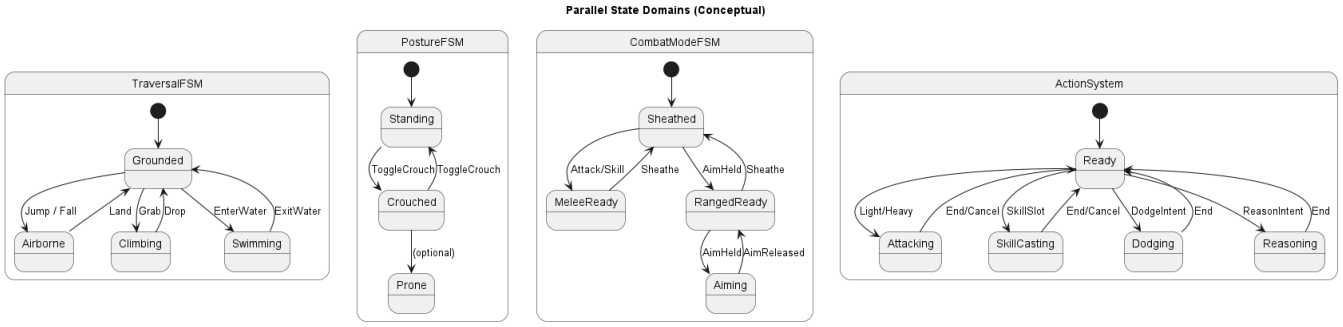



```
@startuml
title Tick Sequence - Input to Outputs

actor Engine as E
participant UnityHost as UH
participant InputAdapter as IA
participant WorldSnapshotBuilder as WSB
participant PlayerPiece as PP
participant IntentResolver as IR
participant InteractionResolver as IX
participant IntentArbiter as ARB
participant Traversal as TR
participant Posture as PO
participant CombatMode as CM
participant ActionSystem as AS
participant OutputBuilder as OB
participant MotorAdapter as MA
participant AnimatorAdapter as AA

E -> UH : Update()
UH -> IA : BuildInputSnapshot()
IA --> UH : InputSnapshot
UH -> WSB : BuildWorldSnapshot()
WSB --> UH : WorldSnapshot

UH -> PP : Tick(InputSnapshot, WorldSnapshot, dt)
PP -> IR : Resolve(raw input + model + world)
IR --> PP : intents (candidates)
PP -> IX : Resolve context intents (Circle etc.)
IX --> PP : refined intents
PP -> ARB : Select/consume/buffer intents
ARB --> PP : intents to execute
```

```
PP -> TR : HandleIntents + Tick
PP -> PO : HandleIntents + Tick
PP -> CM : HandleIntents + Tick
PP -> AS : HandleIntents + Tick (may lock others)

PP -> OB : BuildOutputs()
OB --> PP : CharacterOutputs
PP --> UH : TickResult(outputs, events)

UH -> MA : Apply(MotorCommands)
UH -> AA : Apply(AnimationCommands)

@enduml
```

6.4 Multi-domain state model (conceptual)

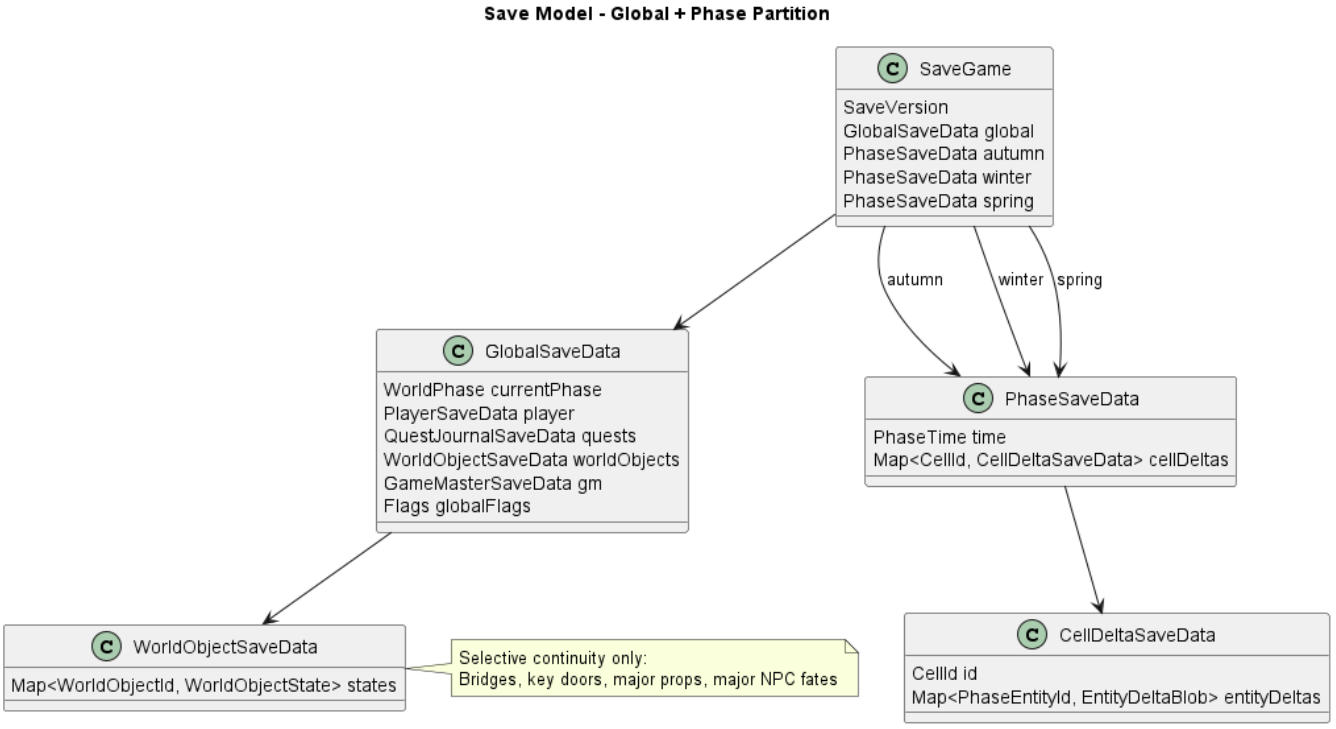
This is not a strict UML state chart (because each domain is separate), but it shows the parallel state concept.




```
@startuml
title Parallel State Domains (Conceptual)

state "TraversalFSM" as T {
  [*] --> Grounded
  Grounded --> Airborne : Jump / Fall
  Airborne --> Grounded : Land
  Grounded --> Climbing : Grab
  Climbing --> Grounded : Drop
  Grounded --> Swimming : EnterWater
  Swimming --> Grounded : ExitWater
}

state "PostureFSM" as P {
  [*] --> Standing
  Standing --> Crouched : ToggleCrouch
  Crouched --> Standing : ToggleCrouch
  Crouched --> Prone : (optional)
}

state "CombatModeFSM" as C {
  [*] --> Sheathed
  Sheathed --> MeleeReady : Attack/Skill
```

```
  Sheathed --> RangedReady : AimHeld
  MeleeReady --> Sheathed : Sheathe
  RangedReady --> Sheathed : Sheathe
  RangedReady --> Aiming : AimHeld
  Aiming --> RangedReady : AimReleased
}

state "ActionSystem" as A {
  [*] --> Ready
  Ready --> Attacking : Light/Heavy
  Ready --> SkillCasting : SkillSlot
  Ready --> Dodging : DodgeIntent
  Ready --> Reasoning : ReasonIntent
  Attacking --> Ready : End/Cancel
  SkillCasting --> Ready : End/Cancel
  Dodging --> Ready : End
  Reasoning --> Ready : End
}

@enduml
```

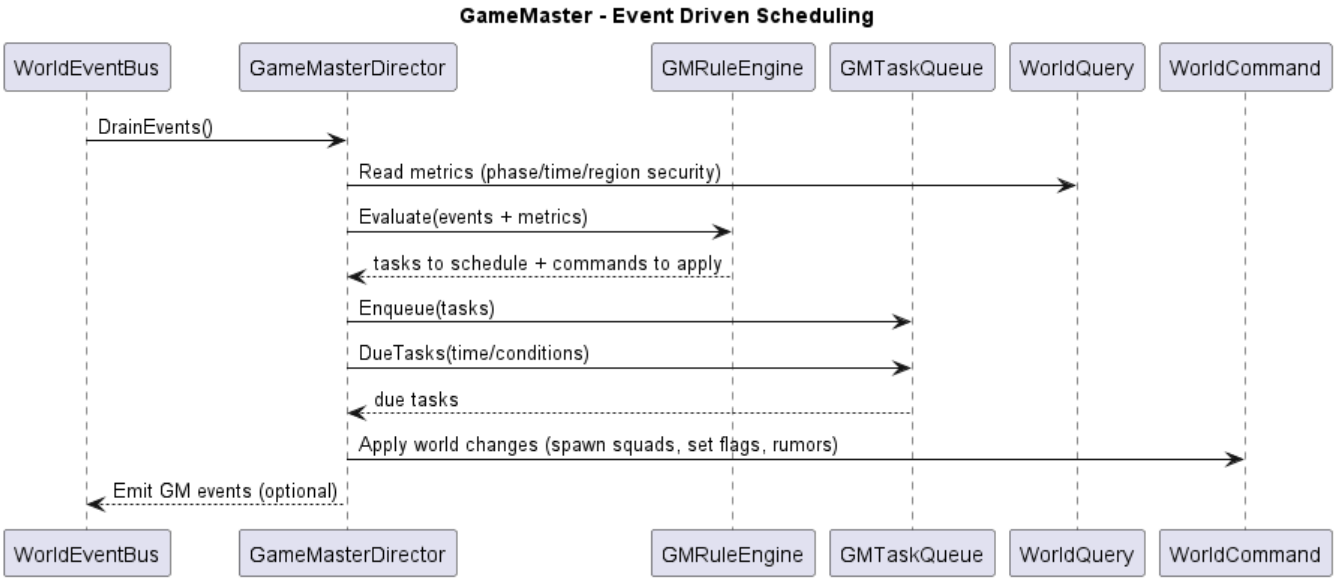
6.5 Persistence partitioning by phase with selective WorldObjectId




```
@startuml
title Save Model - Global + Phase Partition

class SaveGame {
  SaveVersion
  GlobalSaveData global
  PhaseSaveData autumn
  PhaseSaveData winter
```

```
  PhaseSaveData spring
}

class GlobalSaveData {
  WorldPhase currentPhase
  PlayerSaveData player
  QuestJournalSaveData quests
  WorldObjectSaveData worldObjects
  GameMasterSaveData gm
  Flags globalFlags
}

class PhaseSaveData {
  PhaseTime time
  Map<CellId, CellDeltaSaveData> cellDeltas
}

class CellDeltaSaveData {
  CellId id
  Map<PhaseEntityId, EntityDeltaBlob> entityDeltas
}

class WorldObjectSaveData {
  Map<WorldObjectId, WorldObjectState> states
}

SaveGame --> GlobalSaveData
SaveGame --> PhaseSaveData : autumn
SaveGame --> PhaseSaveData : winter
SaveGame --> PhaseSaveData : spring

GlobalSaveData --> WorldObjectSaveData
PhaseSaveData --> CellDeltaSaveData

note right of WorldObjectSaveData
Selective continuity only:
Bridges, key doors, major props, major NPC fates
end note

@enduml
```

6.6 GameMaster director loop




```
@startuml
title GameMaster - Event Driven Scheduling

participant WorldEventBus as WEB
participant GameMasterDirector as GM
participant GMRuleEngine as RE
participant GMTaskQueue as Q
participant WorldQuery as WQ
participant WorldCommand as WC

WEB -> GM : DrainEvents()
GM -> WQ : Read metrics (phase/time/region security)
GM -> RE : Evaluate(events + metrics)
RE --> GM : tasks to schedule + commands to apply
GM -> Q : Enqueue(tasks)
GM -> Q : DueTasks(time/conditions)
Q --> GM : due tasks
GM -> WC : Apply world changes (spawn squads, set flags, rumors)
GM --> WEB : Emit GM events (optional)

@enduml
```

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7. Responsibility Matrix (non-negotiable)

Concern	Owner
Grounded/Airborne/Swimming/Climbing semantics	TraversalFSM
Stance/collider profile, baseline noise multiplier	PostureFSM
Sheathed/MeleeReady/RangedReady, Aiming mode	CombatModeFSM

Concern	Owner
Attacks/skills/dodge/social actions, stamina, buffering, cancels	ActionSystem
Noise/visibility continuous emission	StealthTelemetry
Interaction opportunity detection (pickpocket, takedown, lockpick, reason)	OpportunityDetector + InteractionResolver
Raw input device mapping/chords	Engine Input System + InputAdapter
Contextual meaning of input	IntentResolver + InteractionResolver
World phase baseline content	Content + World loader
Per-phase local deltas	World.Core + Persistence
Cross-phase continuity objects	WorldObjectState (Global)
Macro reactive world	GameMasterDirector

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8. Key Data Types (silhouettes)

8.1 InputSnapshot

- `Move: Vec2`, `Look: Vec2`
- Buttons as `ButtonState`:
  - `Primary`, `Secondary`, `Tertiary`, `Jump`
  - `AimModifier`, `SkillModifier`
  - `ContextGrabOrFire`
  - plus toggles as needed

8.2 WorldSnapshot

- Contact and environment features:
  - `IsGrounded`, `IsInWater`, `CanClimb`, `CanMantle`, `SurfaceTag`
- Nearby interactables summary:
  - best candidate door, NPC, loot, etc.
- Optional: Threat/social context summary

8.3 Intent types

- traversal intents: `JumpIntent`, `GrabOrClimbIntent`, `MantleIntent`
- combat intents: `LightAttackIntent`, `HeavyAttackIntent`
- skill intents: `UseSkillIntent(slot, kind)`
- ranged intents: `FireBasicRangedIntent`

- interaction intents: `LockpickIntent`, `PickpocketIntent`, `TalkIntent`, `ReasonIntent(targetId)`

8.4 Outputs

- `MotorCommands`: move intent, speed cap, root motion enable, align-to-target
- `AnimationCommands`: param changes, triggers, action tags
- `CameraCommands`: aim camera, lock-on suggestions
- `SfxVfxCommands`: footsteps, impact, stealth whoosh, etc.

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9. Save/Load Policies

9.1 Saved

- Player: stats, equipment, skill loadouts, reputation, resources (configurable)
- Quests: journal and objective progress
- World: current phase + per-phase cell deltas + global WorldObject states
- GameMaster: scheduled tasks + metrics + cooldowns

9.2 Not saved (recomputed)

- FSM current state instances
- animation graph runtime
- cached physics queries
- input buffers (unless explicit design requires)
- transient “in combat” flags

9.3 Phase transitions

When phase changes:

- Flush current phase deltas
- Update global phase state
- Run quest transition policies (expire/convert)
- Load phase baseline world + apply phase deltas
- Apply global WorldObjectState

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10. Implementation Roadmap (incremental)

Milestone 1 (clean seam)

- Introduce `InputSnapshot`, `IntentResolver`, and `Tick` pipeline
- Convert existing locomotion FSM to consume intents rather than input events
- Output commands instead of direct Animator/CharacterController calls (adapter applies)

Milestone 2 (combat/action)

- Add CombatMode model (aim + weapon readiness)
- Add ActionSystem for light/heavy/skill/dodge
- Add InteractionResolver for Circle basics

Milestone 3 (stealth/posture/social)

- PostureFSM + StealthTelemetry
- Opportunity detection for takedown/pickpocket/lockpick
- Add `ReasonAction` as a first social action

Milestone 4 (world phases + persistence)

- Implement phase baseline world loading
- Implement per-phase deltas + WorldObjectId global state
- JSON save system with versioning

Milestone 5 (GameMaster)

- Event bus integration
- GM task scheduling and 2–3 reactive rules
- GM persistence

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11. Invariants & Guardrails

1. Core modules must compile without engine references.
2. No system other than the domain owner mutates that domain's state.
3. All contextual input meaning is resolved before domain modules.
4. External systems interact with pieces only via `ICharacterReadModel` and `ICharacterCommandPort`.
5. Per-phase deltas never apply outside their phase.
6. Only selective objects receive cross-phase `WorldObjectId`.